King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 208 Major Exam II Term 251

Original exam date: 11 November 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. The general solution of the differential equation

$$(D^5 - 2D^3 - 2D^2 - 3D - 2)y = 0$$

is

(a) $y(x) = e^{-x}(c_1 + c_2x) + c_3e^{2x} + c_4e^{-x}\cos x + c_5e^{-x}\sin x$

(b) $y(x) = c_1e^{-x} + (c_2 + c_3x)e^{2x} + c_4\cos x + c_5\sin x$

(c) $y(x) = e^{-x}(c_1 + c_2x) + c_3e^{2x} + c_4\cos x + c_5\sin x$

(d) $y(x) = e^{x}(c_1 + c_2x) + c_3e^{2x} + c_4\cos x + c_5\sin x$

(e) $y(x) = e^{-x}(c_1 + c_2x) + c_3e^{2x} + c_4e^{x}\cos x + c_5e^{x}\sin x$

2. Given that $y = e^{-x}\cos x$ is a solution of the differential equation $9y^{(3)} + 11y'' + 4y' - 14y = 0$, the general solution of the differential equation is

(a) $y = c_1e^{-\frac{2}{9}x} + c_2e^{-x}\cos x + c_3e^{-x}\sin x$

(b) $y = c_1e^{-\frac{7}{9}x} + c_2e^{-x}\cos x + c_3e^{-x}\sin x$

(c) $y = c_1e^{\frac{7}{9}x} + c_2e^{-x}\cos x + c_3e^{-x}\sin x$

(d) $y = c_1e^{\frac{2}{9}x} + c_2e^{-x}\cos x + c_3e^{-x}\sin x$

(e) $y = c_1e^{x} + c_2e^{-x}\cos x + c_3e^{-x}\sin x$

3. If the solution space of the system

$$\begin{aligned} x_1 - 2x_2 - 3x_3 - 16x_4 &= 0 \\ 2x_1 - 4x_2 + x_3 + 17x_4 &= 0 \\ x_1 - 2x_2 + 3x_3 + 26x_4 &= 0 \end{aligned}$$

consists of all linear combinations of the vectors $v_1 = (a, 1, 0, b)$ and $v_2 = (c, 0, -7, d)$, then $a + b + c + d =$

(a) -4

(b) 0

(c) -3

(d) 5

(e) -2

4. If $W(x)$ is the Wronskian of the functions

$$f_1(x) = 2x + 3\cos x, \quad f_2(x) = 5\cos x, \quad f_3(x) = -3x,$$

then $W(x) =$

(a) $-2x$

(b) 0

(c) $\sin x$

(d) $3\cos x$

(e) $2x$

5. If $y(x)$ is the solution of the initial-value problem

$$(D+2)^2 y = 0, \quad y(0) = 1, \quad y'(0) = -1,$$

then $y\left(\frac{1}{2}\right) =$

(a) $\frac{5}{e}$

(b) 0

(c) $\frac{5}{2e}$

(d) $\frac{3}{e}$

(e) $\frac{3}{2e}$

6. An eigenvector associated with the eigenvalue $\lambda = 5$ of the matrix $A = \begin{bmatrix} 5 & 0 \\ 4 & -1 \end{bmatrix}$ is $\begin{bmatrix} 3 \\ \alpha \end{bmatrix}$ where $\alpha =$

(a) 2

(b) 3

(c) -2

(d) 1

(e) -3

7. If the rank of the matrix

$$\begin{bmatrix} 1 & 1 & 3 & 3 & 0 \\ -1 & 0 & -2 & -1 & 1 \\ 2 & 3 & 7 & 8 & \alpha \\ -2 & 4 & 0 & 6 & 7 \end{bmatrix}$$

is equal to 3, then $\alpha =$

(a) 1

(b) 2

(c) 0

(d) -2

(e) 3

8. If the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & 3 \\ 1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$

(c) $P = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

9. A basis for the subspace of $\mathbb{R}^3$ consists of the line of intersection of the planes $x - 4y + 7z = 0$ and $y = -z$ is given by $v = (\alpha, -1, 1)$ where $\alpha =$

(a) 10

(b) 0

(c) -11

(d) 9

(e) -10

10. A linear homogeneous differential equation with real coefficients having the solutions xe^x, $5e^x \cos(3x)$ is

(a) $y^{(4)} + 4y^{(3)} + 15y'' - 22y' + 2y = 0$

(b) $y^{(4)} - 10y^{(3)} + 15y'' - 22y' + 16y = 0$

(c) $y^{(4)} - 4y^{(3)} + 15y'' - 22y' + 10y = 0$

(d) $y^{(4)} - 4y^{(3)} + 13y'' - 22y' + 12y = 0$

(e) $y^{(4)} - 4y^{(3)} + 15y'' - 20y' + 8y = 0$

11. If the characteristic polynomial of the matrix $A = \begin{bmatrix} 0 & 3 & 1 \\ 1 & 2 & 1 \\ 1 & 3 & 0 \end{bmatrix}$ is $p(\lambda) = -(\lambda+1)^2(\lambda-4)$, then a basis for the eigenspace of $\lambda = -1$ is $v_1 = \begin{bmatrix} \alpha \\ 0 \\ -1 \end{bmatrix}$, $v_2 = \begin{bmatrix} \beta \\ -1 \\ 0 \end{bmatrix}$, where $\alpha + \beta =$

(a) 2

(b) 4

(c) -4

(d) 3

(e) 0

12. If $y_p = Ax^3 + Bx^2 + Cx + D\cos x + E\sin x$ is a particular solution of the differential equation

$$y''' + 4y' = 24\left(x^2 + \sin x\right),$$

then $A + B + C + D + E =$

(a) 8

(b) 11

(c) -8

(d) 9

(e) -9

13. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$(D^2 - 1)^2 y = e^x + \sin x$$

is given by $y_p(x) =$

(a) $Ae^x + B\cos x + C\sin x$

(b) $Ax^2e^x + Bx^2\cos x + Cx^2\sin x$

(c) $Ax^2e^x + B\cos x + C\sin x$

(d) $Ax^2e^x + B\sin x$

(e) $Axe^x + B\cos x + C\sin x$

14. A particular solution of the differential equation

$$y'' - 3y' + 2y = \cos\left(e^{-x}\right)$$

is given by $y_p(x) =$

(a) $e^x \cos\left(e^{-x}\right) + e^{2x} \sin\left(e^{-x}\right)$

(b) $-e^{2x} \cos\left(e^{-x}\right)$

(c) $2e^x \cos\left(e^{-x}\right) + e^{2x} \sin\left(e^{-x}\right)$

(d) $-2e^{2x} \cos\left(e^{-x}\right)$

(e) $2e^{2x} \sin\left(e^{-x}\right)$

15. The characteristic equation of the matrix $\begin{bmatrix} 0 & 1 & -1 \\ -3 & 1 & -3 \\ 1 & -4 & 2 \end{bmatrix}$ is

(a) $\lambda^3 - 3\lambda^2 - 6\lambda + 8 = 0$

(b) $\lambda^3 - 3\lambda^2 - 5\lambda + 6 = 0$

(c) $\lambda^3 - 3\lambda^2 + 6\lambda - 8 = 0$

(d) $\lambda^3 - 4\lambda^2 - 6\lambda + 8 = 0$

(e) $\lambda^3 - 5\lambda^2 - 6\lambda - 8 = 0$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(c)**	$(r-2)(r+1)^2(r^2+1)$: $e^{-x}(c_1+c_2x)+c_3e^{2x}+c_4\cos x+c_5\sin x$.
2	**(c)**	Factor r^2+2r+2; $9r^3+11r^2+4r-14=(r^2+2r+2)(9r-7)$, third root $\frac{7}{9}$.
3	**(e)**	RREF: $x_1=2x_2-5x_4$, $x_3=-7x_4$; $v_1=(2,1,0,0)$, $v_2=(-5,0,-7,1)$, sum -2.
4	**(b)**	$f_1=\frac{3}{5}f_2-\frac{2}{3}f_3$: linearly dependent, so $W=0$.
5	**(e)**	Double root -2: $y=(1+x)e^{-2x}$, so $y(\frac{1}{2})=\frac{3}{2e}$.
6	**(a)**	$(A-5I)\mathbf{v}=\mathbf{0}$: $4v_1=6v_2$, so $\mathbf{v}=(3,2)$ and $\alpha=2$.
7	**(a)**	Row reduce: last rows $[0,0,0,0,\alpha-1]$ and $[0,0,0,0,1]$; the α-row vanishes at $\alpha=1$.
8	**(a)**	$\lambda=1,2$, eigenvectors $(3,2)$, $(1,1)$: $P=\begin{bmatrix}3 & 1\\ 2 & 1\end{bmatrix}$, $D=\mathrm{diag}(1,2)$.
9	**(c)**	$y=-z$ gives $x=-11z$, so $v=(-11,-1,1)$.
10	**(c)**	Roots $1,1,1\pm 3i$: $(r-1)^2((r-1)^2+9)=r^4-4r^3+15r^2-22r+10$.
11	**(b)**	$A+I$ gives $x+3y+z=0$: $v_1=(1,0,-1)$, $v_2=(3,-1,0)$, so $\alpha+\beta=4$.
12	**(e)**	Matching gives $A=2, B=0, C=-3, D=-8, E=0$, sum -9.
13	**(c)**	Double root $r=1$ gives Ax^2e^x; $\pm i$ not roots gives $B\cos x+C\sin x$.
14	**(b)**	Variation of parameters, $y_1=e^x$, $y_2=e^{2x}$, $W=e^{3x}$: $y_p=-e^{2x}\cos(e^{-x})$.
15	**(a)**	$\det(A-\lambda I)=-\lambda^3+3\lambda^2+6\lambda-8$, so $\lambda^3-3\lambda^2-6\lambda+8=0$.